

Introduction

Introduction to Programming

StayAhead Training; Sep, 2024

Thinking Like a Computer

Introduction

- What can a computer do?

Thinking Like a Computer

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- What can a computer do?
 1. Input and output (I/O)

Thinking Like a Computer

Introduction

- What can a computer do?
 1. Input and output (I/O)
 2. Storage

Thinking Like a Computer

Introduction

- What can a computer do?
 1. Input and output (I/O)
 2. Storage
 3. Arithmetic

Thinking Like a Computer

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- What can a computer do?
 1. Input and output (I/O)
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 3. Arithmetic
 4. Comparisons

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- What can a computer do?
 1. Input and output (I/O)
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 4. Comparisons
 5. Decisions

Thinking Like a Computer

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- What can a computer do?
 1. Input and output (I/O)
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 5. Decisions
 6. Repetition

Thinking Like a Computer

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- What can a computer do?
 1. Input and output (I/O)
 2. Storage
 3. Arithmetic
 4. Comparisons
 5. Decisions
 6. Repetition
 7. Reuse

What Constitutes Data?

Introduction

- 0 (off) and 1 (on)
- 1000001 (65)
- 65 (A)

Data Jargon

Introduction

Jargon	Description
Byte	A binary number comprising eight bits
Kilobyte (kB)	A thousand bytes
Megabyte (MB)	A thousand kilobytes
Gigabyte (GB)	A thousand megabytes
ASCII	Character encoding system (English only; 1 byte/character)
UTF-8	Character encoding system (All languages; 1-4 bytes/character)

What is a Program?

Introduction

- **Program:** set of instructions to be executed by a computer
- **Instruction:** I/O, storage, arithmetic, comparison, etc.
- **Statement:** instruction (one or more lines of code)
- **Comment:** note about the code (ignored by the computer)

What is Code?

Introduction

- **Code:** system of words, symbols and rules
- **Programming language:** specific system of words, symbols and rules

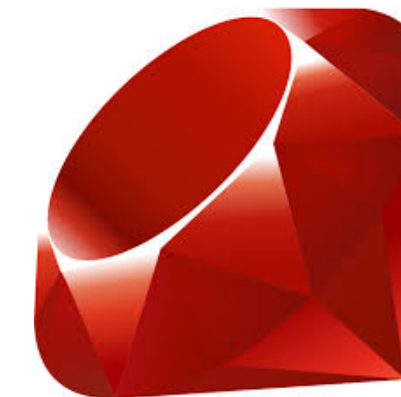
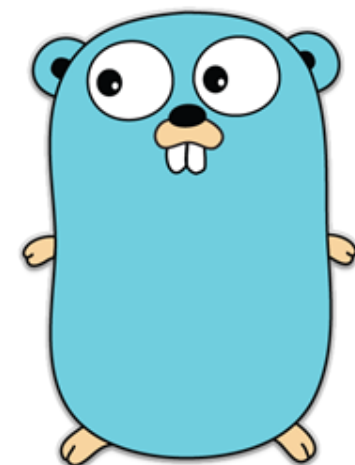
What is Code?

Introduction

- PY: `print("code")`
- `_____`
- JS: `console.log("code");`
- `_____`
- JV: `System.out.println("code");`
- `_____`
- C#: `Console.WriteLine("code");`

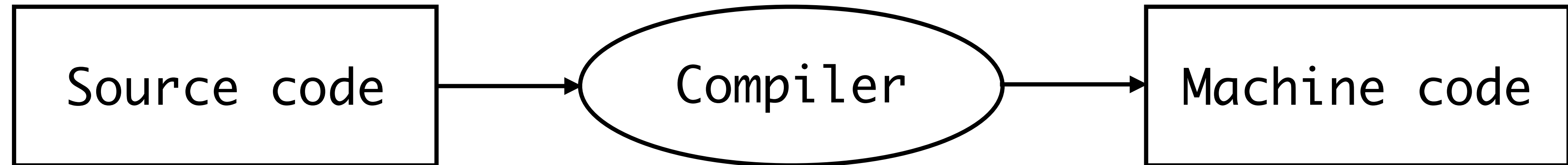
Why So Many Programming Languages?

Introduction



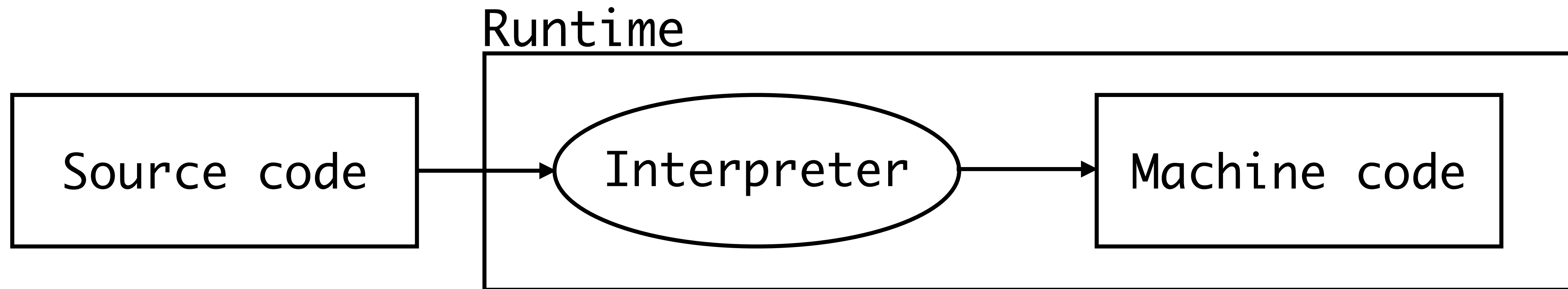
From Source Code to Runtime

Introduction



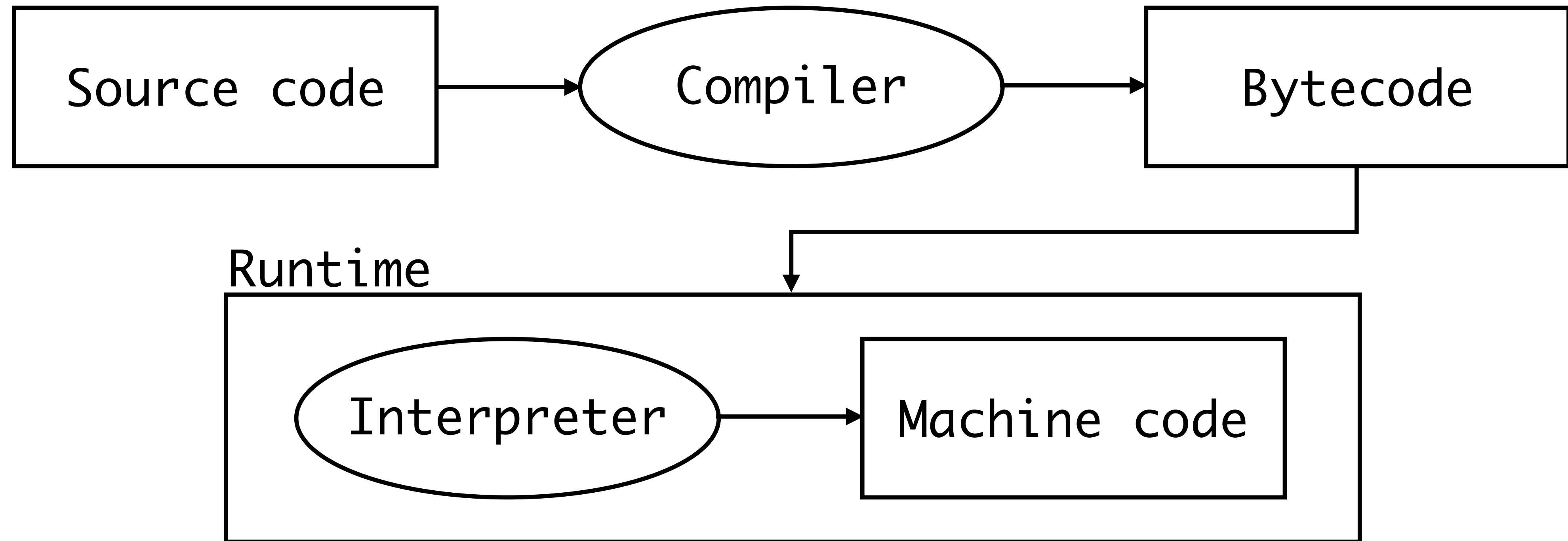
From Source Code to Runtime

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From Source Code to Runtime

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What Does a Programmer Do?

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- What does a programmer do?

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 1. Problem specification

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 4. Testing

Getting Ready to Code

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- Which language are we going to use?
- REPL shell
- Standard input (stdin) and output (stdout)